

SUPPLEMENTARY MATERIAL: PRECISION ACCOUNTING FOR HYBRID LLM CACHES

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Table 1. Architecture-derived continuous-model parameters. All values are bytes. No parameter is fitted to allocator token counts.

Model	A_f	A_q	G_{fp32}	G_{bf16}
2B	12,288	3,168	19,537,920	10,100,736
9B	16,384	4,224	26,050,560	13,467,648

1 BYTE-EXACT PARAMETER DERIVATION

The attention term includes all GQA layers. At fp16, one attention layer uses two tensors (K and V), 512 elements per token, and 2 bytes per element. This is 2,048 bytes per layer and token. At int4, K and V each use 256 packed payload bytes plus 8 metadata bytes. The total is therefore $2 \times (256 + 8) = 528$ bytes per layer and token.

Each GDN layer contains two state components. The convolution state is bf16 in both configurations and uses 36,864 bytes. The temporal state uses 1,048,576 bytes at fp32 or 524,288 bytes at bf16. Consequently,

$$G_{fp32}^{\text{layer}} = 36,864 + 1,048,576 = 1,085,440, \quad (1)$$

$$G_{bf16}^{\text{layer}} = 36,864 + 524,288 = 561,152. \quad (2)$$

The complete layer state falls by 48.30%, not 50%.

The 2B model has six GQA and 18 GDN layers. The 9B model has eight GQA and 24 GDN layers. Multiplying the per-layer values yields Table 1. The representative padded recurrent pages are 1,089,792 bytes at fp32 and 566,016 bytes at bf16. Padding therefore adds 4,352 and 4,864 bytes, respectively. The continuous model uses unpadded architecture bytes; the discrete allocator uses the padded layout.

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2 DISCRETE ALLOCATOR RECONSTRUCTION

Let K be the GPU-block count, H the recurrent cache-group count, B the attention block size, and L the requested context. The allocator-equivalent slot count and token capacity are

$$S_{\text{alloc}} = \frac{K}{H + \lceil L/B \rceil}, \quad T_{\text{alloc}} = \lfloor LS_{\text{alloc}} \rfloor. \quad (3)$$

For 2B/int4/4K with fp32 state, $K = 3287$, $H = 3$, and $B = 2064$. The denominator is $3 + \lceil 4096/2064 \rceil = 5$. Thus $S_{\text{alloc}} = 657.4$ and $T_{\text{alloc}} = 2,692,710$. With bf16 state, $K = 6330$ and $B = 1072$. The denominator is seven, $S_{\text{alloc}} = 904.286$, and $T_{\text{alloc}} = 3,703,954$.

Fractional slot counts are an allocator normalization. They do not mean that the scheduler admits a fractional request, nor do they prove admission of the integer floor under prefill, decode, or latency constraints.

3 FULL-MATRIX AND TEMPORAL CHECKS

The formal matrix contains 112 allocator cells and 52 matched fp32/bf16 core pairs. Every pair favors bf16 state. The median capacity increase is 15.44%, with a range of 3.32–93.28%. The corrected continuous-model residual has 2.3795% median absolute magnitude, 2.7955% mean absolute magnitude, and 13.2134% maximum absolute magnitude. Signed residuals span -5.1604% to $+13.2134\%$.

The temporal rerun repeats the frozen matrix. The allocator is deterministic for a fixed build and configuration, so this is a reproducibility check rather than an independent statistical replicate. All cell directions agree and the maximum token difference is 1.42%.

4 DEPENDENCE-AWARE GSM8K ANALYSIS

Each dataset seed samples 200 GSM8K items without replacement. Samples overlap across seeds, producing 1,800 seed-item draws but only 1,017 unique items. Pairwise seed overlap ranges from 19 to 39, with median 30. The primary estimand is the mean paired difference over the observed seed-item draws.

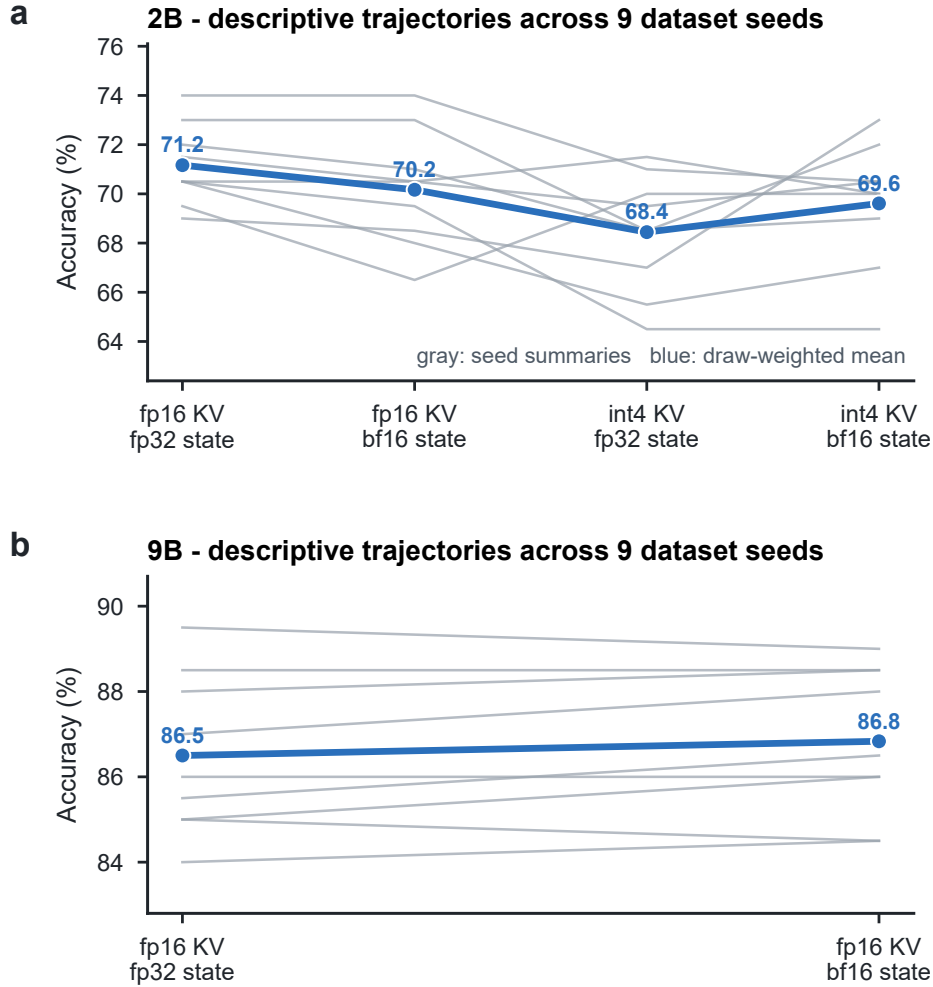


Figure 1. Descriptive GSM8K dataset-seed summaries. Thin lines show nine seed summaries. The blue line is the draw-weighted mean. These lines are not independent replicates and are not the basis of the confidence intervals.

For each candidate, an intercept-only OLS model is fit to paired binary differences. The covariance uses two-way CR1 clustering by item and dataset seed. The reference distribution is Student- t with $\min(1017 - 1, 9 - 1) = 8$ degrees of freedom. A percentile bootstrap samples the 1,017 item-level means with equal item weight for 20,000 replicates.

Repeated items are not always deterministic across dataset seeds despite greedy decoding. The 2B full, state-only, KV-only, and joint allocations have 30, 27, 47, and 43 repeated items with inconsistent outcomes. Clustering accounts for dependence but does not explain this behavior. The old script that treated seed means as iid now exits with a deprecation error.

5 ADDITIONAL EVIDENCE PANELS

6 ARTIFACT AND VERIFICATION MANIFEST

The current anonymous draft does not claim a public artifact release. The repository-local verifier checks SHA-256 digests for 16 frozen sources, asserts headline values and all 52 paired capacity rows, and rejects old capacity and seed-iid GSM8K inputs. It also checks all eight SVG/PDF pairs for nonempty vector output. The selector audit generator verifies the four decision sidecars before producing its JSON and LaTeX tables.

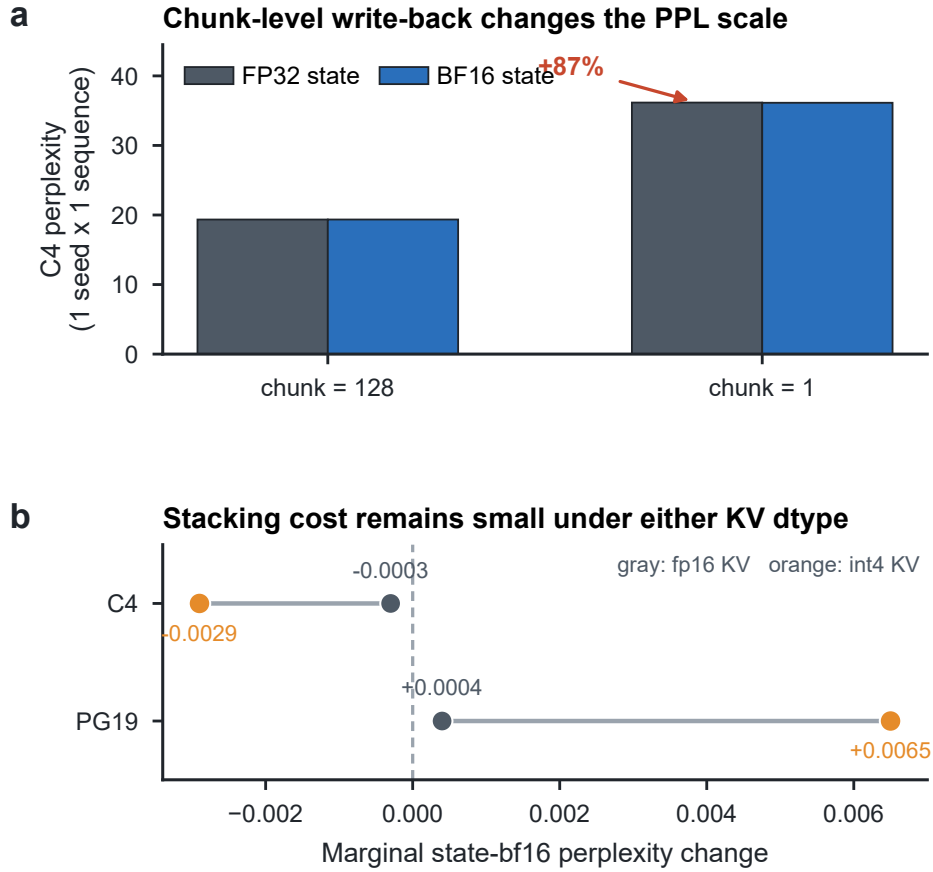


Figure 2. PPL harness boundary. (a) Reducing chunk size from 128 to one changes the PPL scale by about 87%, so the harness is not equivalent to per-token kernel semantics. (b) State contrasts under fp16 and int4 KV have wide three-seed uncertainty and do not establish equivalence.

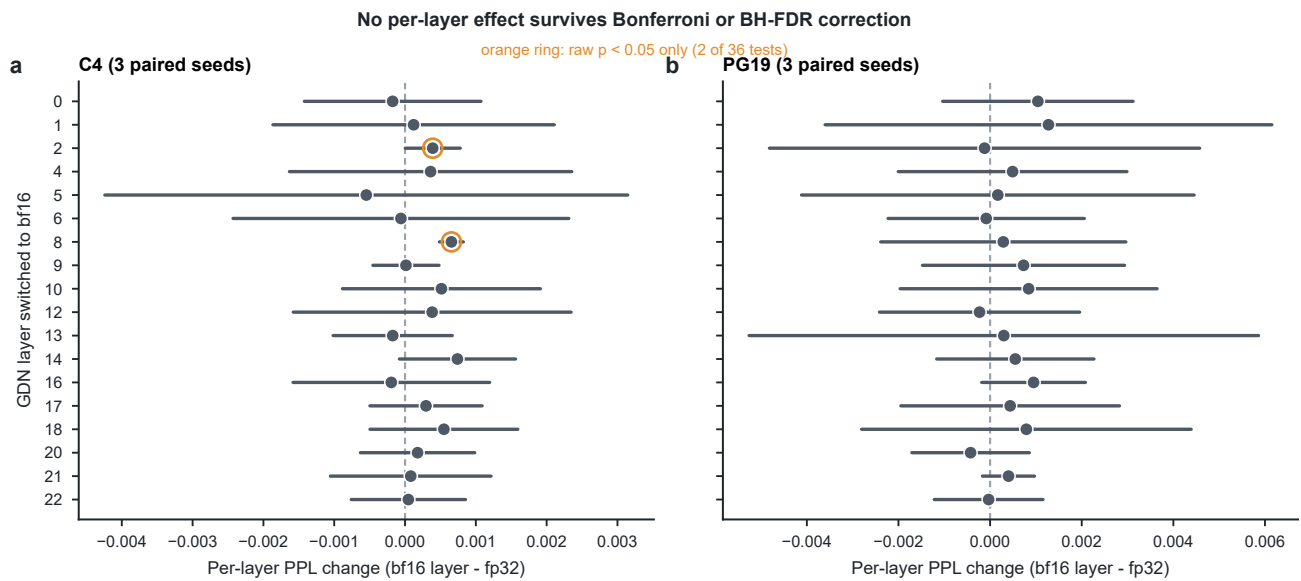


Figure 3. Exploratory per-layer sensitivity. Two of 36 raw tests have $p < 0.05$. None survives Bonferroni or Benjamini–Hochberg correction. This screen does not establish that whole-state precision is optimal.

7 COMPLETE PAIRED ALLOCATOR MATRIX

Table 2 enumerates every fp32/bf16 pair in the formal 112-cell run. A pair is uniquely identified by model, KV dtype, context length, and GPU-memory utilization. The remaining eight cells probe a float16-state frontier and are not part of the 52-pair state comparison. The frozen source is `capacity-phase-formal-corrected.analysis.json`; its SHA-256 prefix is 879dc059579e, with the full digest in the adjacent sidecar.

Table 2: **Complete formal allocator matrix.** K is the allocated GPU-block count and T is total allocator token capacity. Gain is $100(T_{16}/T_{32} - 1)$; residual is the signed measured-to-continuous state-ratio error. The 52 rows come from attempt `capacity-phase-formal-20260811`; they are allocator measurements, not scheduler-admitted concurrency.

Model	KV	L	u	K_{32}	K_{16}	T_{32}	T_{16}	Gain	Resid.
2B	fp16	1,024	0.70	2,466	4,658	505,036	681,398	34.92%	-4.72%
2B	fp16	1,024	0.80	2,969	5,609	608,051	820,516	34.94%	-4.70%
2B	fp16	1,024	0.90	3,473	6,561	711,270	959,780	34.94%	-4.71%
2B	fp16	2,048	0.70	2,465	4,657	721,188	867,048	20.22%	-5.16%
2B	fp16	2,048	0.80	2,969	5,609	868,644	1,044,293	20.22%	-5.16%
2B	fp16	2,048	0.90	3,473	6,561	1,016,100	1,221,538	20.22%	-5.16%
2B	fp16	4,096	0.70	2,466	4,658	918,248	1,059,953	15.43%	-0.16%
2B	fp16	4,096	0.80	2,969	5,609	1,105,547	1,276,359	15.45%	-0.14%
2B	fp16	4,096	0.90	3,473	6,561	1,293,218	1,492,992	15.45%	-0.15%
2B	fp16	8,192	0.70	2,465	4,657	1,062,804	1,192,192	12.17%	+3.37%
2B	fp16	8,192	0.80	2,969	5,609	1,280,107	1,435,904	12.17%	+3.36%
2B	fp16	8,192	0.90	3,473	6,561	1,497,411	1,679,616	12.17%	+3.36%
2B	fp16	16,384	0.70	2,466	4,658	1,188,321	1,271,944	7.04%	+2.46%
2B	fp16	16,384	0.80	2,969	5,609	1,430,708	1,531,630	7.05%	+2.48%
2B	fp16	16,384	0.90	3,473	6,561	1,673,577	1,791,590	7.05%	+2.48%
2B	fp16	32,768	0.70	2,466	4,658	1,262,592	1,304,558	3.32%	+1.01%
2B	fp16	32,768	0.80	2,969	5,609	1,520,128	1,570,903	3.34%	+1.03%
2B	fp16	32,768	0.90	3,473	6,561	1,778,176	1,837,528	3.34%	+1.03%
2B	int4	1,024	0.70	2,515	4,843	643,840	1,239,808	92.56%	+12.80%
2B	int4	1,024	0.80	3,019	5,835	772,864	1,493,760	93.28%	+13.21%
2B	int4	1,024	0.90	3,545	6,826	907,520	1,747,456	92.55%	+12.79%
2B	int4	2,048	0.70	2,515	4,843	1,287,680	1,983,692	54.05%	-1.81%
2B	int4	2,048	0.80	3,030	5,835	1,551,360	2,390,016	54.06%	-1.80%
2B	int4	2,048	0.90	3,545	6,826	1,815,040	2,795,929	54.04%	-1.81%
2B	int4	4,096	0.70	2,515	4,843	2,060,288	2,833,846	37.55%	-2.38%
2B	int4	4,096	0.80	3,030	5,835	2,482,176	3,414,308	37.55%	-2.37%
2B	int4	4,096	0.90	3,545	6,826	2,904,064	3,994,185	37.54%	-2.38%
2B	int4	8,192	0.70	2,515	4,843	2,943,268	3,606,714	22.54%	-2.88%
2B	int4	8,192	0.80	3,030	5,835	3,545,965	4,345,483	22.55%	-2.88%
2B	int4	8,192	0.90	3,545	6,826	4,148,662	5,083,508	22.53%	-2.89%
2B	int4	16,384	0.70	2,515	4,822	3,745,978	4,158,086	11.00%	-3.66%
2B	int4	16,384	0.80	3,030	5,835	4,513,047	5,031,612	11.49%	-3.24%
2B	int4	16,384	0.90	3,545	6,805	5,280,116	5,868,058	11.14%	-3.55%
2B	int4	32,768	0.70	2,515	4,843	4,337,448	4,667,512	7.61%	-0.62%
2B	int4	32,768	0.80	3,030	5,835	5,225,633	5,623,567	7.62%	-0.62%
2B	int4	32,768	0.90	3,545	6,826	6,113,818	6,578,657	7.60%	-0.63%
9B	fp16	2,048	0.80	290	562	84,845	104,634	23.32%	-2.71%
9B	fp16	2,048	0.90	484	940	141,604	175,010	23.59%	-2.50%
9B	fp16	4,096	0.80	290	563	107,985	121,370	12.40%	-2.79%
9B	fp16	4,096	0.90	484	940	180,224	202,644	12.44%	-2.75%
9B	fp16	16,384	0.80	290	562	135,753	143,872	5.98%	+1.45%
9B	fp16	16,384	0.90	484	940	226,567	240,640	6.21%	+1.67%
9B	fp16	32,768	0.80	288	560	142,987	147,984	3.49%	+1.18%
9B	fp16	32,768	0.90	483	938	239,802	247,874	3.37%	+1.06%
9B	int4	2,048	0.80	287	567	146,944	232,243	58.05%	+0.74%
9B	int4	2,048	0.90	482	942	246,784	385,843	56.35%	-0.34%
9B	int4	4,096	0.80	288	559	235,929	327,094	38.64%	-1.60%
9B	int4	4,096	0.90	482	950	394,854	555,885	40.78%	-0.08%
9B	int4	16,384	0.80	288	567	428,962	488,933	13.98%	-1.08%
9B	int4	16,384	0.90	478	950	711,959	819,200	15.06%	-0.14%
9B	int4	32,768	0.80	286	564	493,244	528,032	7.05%	-1.14%
9B	int4	32,768	0.90	481	948	829,547	887,544	6.99%	-1.19%